

Insert new annexes Annex 162A through Annex 162D corresponding to Clause 162 as follows:

Annex 162A

(informative)

Transmitter, receiver and channel parameters associated with test points TP0 and TP5 for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4

162A.1 Overview

This annex provides information on transmitter, receiver, and channel parameters associated with test points TP0 and TP5, which might not be testable in an implemented system. TP0 and TP5 test points are illustrated in the 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 link block diagram of Figure 162–2. This annex also provides information on channel characteristics.

162A.2 Transmitter characteristics at TP0

The recommended transmitter characteristics at TP0 as measured at TP0v are described in 163.9.2.

162A.3 Receiver characteristics at TP5

The recommended receiver characteristics at TP5 as measured at TP5v are described in 163.9.3.

162A.4 Transmitter and receiver differential printed circuit board (PCB) trace insertion loss

The recommended maximum and minimum insertion loss allocation for the transmitter or receiver differential controlled impedance PCBs are determined using Equation (162A–1) and Equation (162A–2), respectively, and illustrated in Figure 162A–1. Note that the recommended maximum insertion loss allocation for the transmitter or receiver differential controlled impedance PCBs with allowances for ball grid array (BGA) footprint and host connector footprints is 6.875 dB at 26.56 GHz and the recommended minimum insertion loss allocation for the transmitter or receiver differential controlled impedance PCBs is 2.3 dB at 26.56 GHz.

$$ILdd_{PCB}(f) \leq ILdd_{PCBmax}(f) = 0.9809(0.471\sqrt{f} + 0.1194f + 0.002f^2) \quad (162A-1)$$

for 0.01 GHz $\leq f \leq$ 50 GHz

where

$ILdd_{PCB}(f)$	is the insertion loss of the transmitter and receiver PCB in dB
$ILdd_{PCBmax}(f)$	is the maximum insertion loss of the transmitter and receiver PCB in dB
f	is the frequency in GHz

$$ILdd_{PCB}(f) \geq ILdd_{PCBmin}(f) = 0.0059 + 0.0361\sqrt{f} + 0.0794f \quad (162A-2)$$

for 0.01 GHz $\leq f \leq$ 50 GHz

where

$ILdd_{PCB}(f)$ is the insertion loss of the transmitter and receiver PCB in dB
 $ILdd_{PCBmin}(f)$ is the minimum insertion loss of the transmitter and receiver PCB in dB
 f is the frequency in GHz

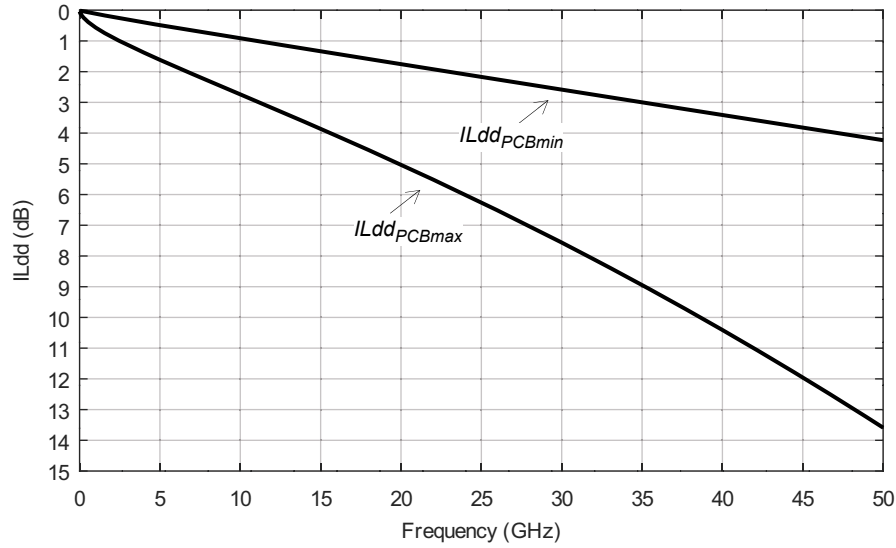


Figure 162A-1—Maximum and minimum insertion loss for transmitter or receiver PCB

The recommended maximum insertion loss from TP0 to TP2 or from TP3 to TP5 including the test fixture is determined using Equation (162A-3), and illustrated in Figure 162A-2. The recommended maximum insertion loss allocation for the transmitter or receiver differential controlled impedance PCBs is consistent with the insertion loss from TP0 to TP2 or TP3 to TP5 and an assumed mated connector insertion loss of 1.6 dB. Note that the recommended maximum insertion loss from TP0 to TP2 or from TP3 to TP5 is 10.975 dB at 26.56 GHz.

$$ILdd_{Host}(f) \leq ILdd_{HostMax}(f) = 1.5658(0.471\sqrt{f} + 0.1194f + 0.002f^2) \quad (162A-3)$$

for $0.01 \text{ GHz} \leq f \leq 50 \text{ GHz}$

where

$ILdd_{Host}(f)$ is the insertion loss from TP0 to TP2 or from TP3 to TP5 including the test fixture in dB

$ILdd_{HostMax}(f)$ is the recommended maximum insertion loss from TP0 to TP2 or from TP3 to TP5 including the test fixture in dB

f is the frequency in GHz

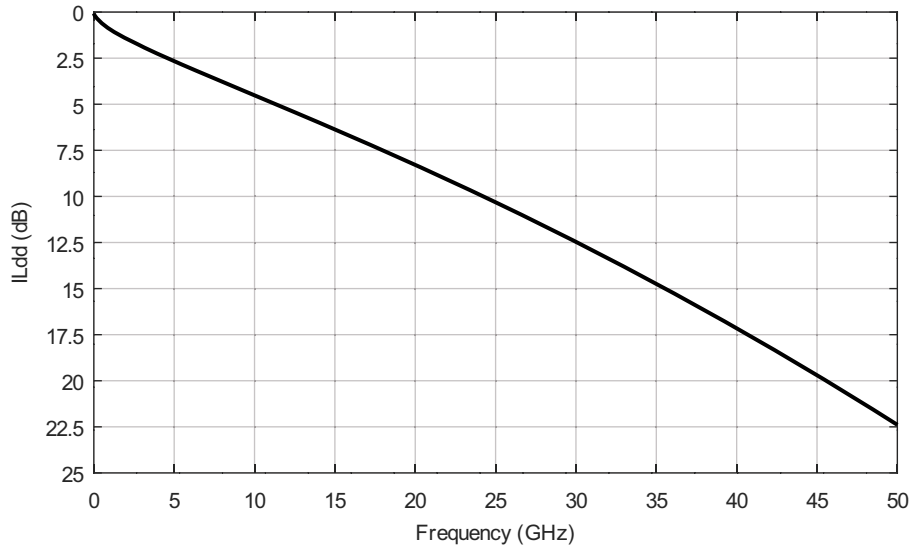


Figure 162A-2—Insertion loss from TP0 to TP2 or from TP3 to TP5

162A.5 Channel insertion loss

This subclause provides information on channel insertion losses for the maximum-loss and minimum-loss cabling topologies.

The maximum channel insertion loss associated with the maximum loss cable assembly is determined using Equation (162A-4). The channel insertion loss budget at 26.56 GHz is illustrated in Figure 162A-3 and Figure 162A-4. Loss budget parameters for all cable topologies are provided in Table 162A-1.

$$ILdd_{Chmax}(f) = ILdd_{Camax}(f) + 2ILdd_{MaxHost}(f) - 2ILdd_{MTFref}(f) \quad (162A-4)$$

for $0.05 \leq f \leq 40$

where

$ILdd_{Chmax}(f)$ is the maximum channel insertion loss in dB between TP0 and TP5
 $ILdd_{Camax}(f)$ is the maximum cable assembly insertion loss in dB (TP1 to TP4)
 $ILdd_{MaxHost}(f)$ is the maximum insertion loss in dB from TP0 to TP2 or TP3 to TP5 using Equation (162A-3)
 $ILdd_{MTFref}(f)$ is the reference insertion loss in dB of the mated test fixture using Equation (162B-5)
 f is the frequency in GHz

The channel insertion loss associated with the minimum insertion loss cable assembly and a maximum insertion loss host channel is determined by Equation (162A-5).

$$ILdd_{Chmin}(f) = ILdd_{Camin}(f) + 2ILdd_{MaxHost}(f) - 2ILdd_{MTFref}(f) \quad (162A-5)$$

for $0.05 \leq f \leq 40$

where

$ILdd_{Chmin}(f)$ is the channel insertion loss in dB between TP0 and TP5 representative of a minimum insertion loss cable assembly and a maximum loss host channel
 $ILdd_{Camin}(f)$ is the minimum cable assembly insertion loss in dB (TP1 to TP4) given in Equation (162-19) and illustrated in Figure 162-8
 $ILdd_{MaxHost}(f)$ is the maximum insertion loss in dB from TP0 to TP2 or TP3 to TP5 using Equation (162A-3)
 $ILdd_{MTFref}(f)$ is the reference insertion loss in dB of the mated test fixture using Equation (162B-5)
 f is the frequency in GHz

Table 162A-1—Insertion loss budget values at 26.56 GHz

Parameter	Value	Units
$ILdd_{Chmax}$	28.5	dB
$ILdd_{Camax}$	19.75	dB
$ILdd_{Chmin}$	19.75	dB
$ILdd_{Camin}$	11.0	dB
$ILdd_{MaxHost}$	10.975	dB
$ILdd_{MTFref}$	6.6	dB

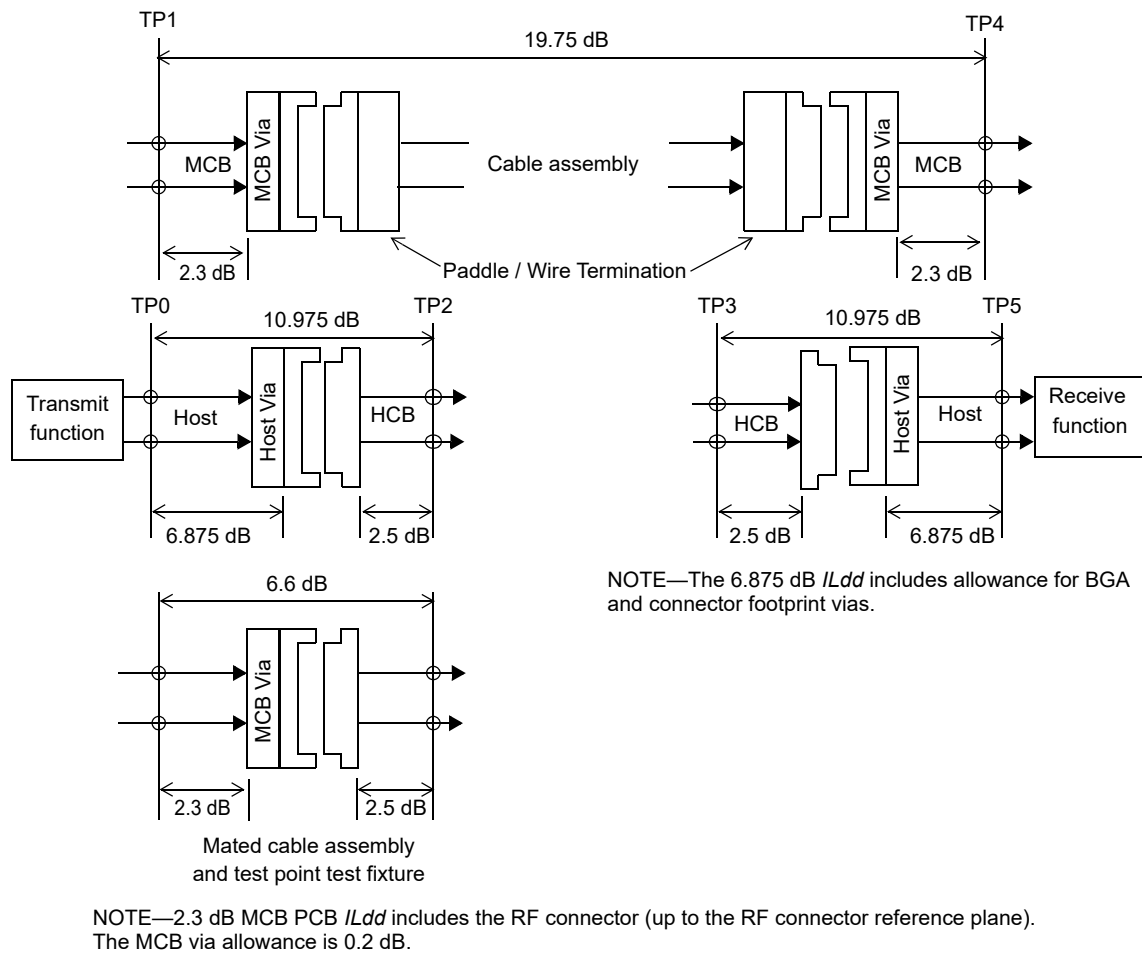
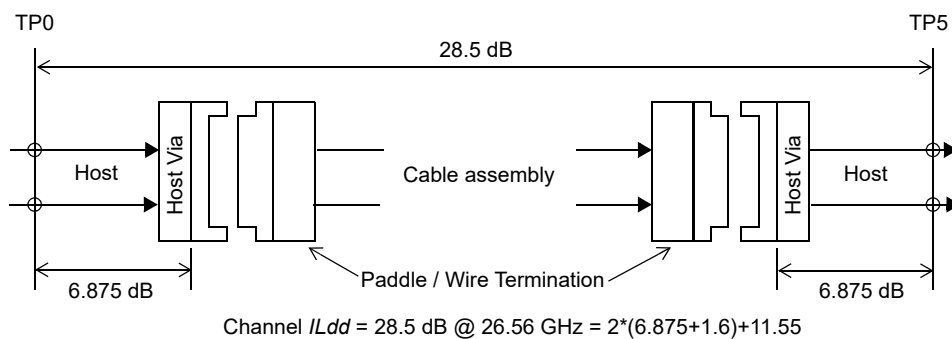


Figure 162A-3—Cable assembly, host, and test fixture insertion loss at 26.56 GHz



NOTE—Channel *ILdd* derived from cable assembly host, and mated test fixture

Figure 162A-4—Channel insertion loss at 26.56 GHz

162A.6 Channel effective return loss (ERL)

The 100GBASE-CR1, 200GBASE-CR2 and 400GBASE-CR4 channels are recommended to meet the ERL specified in 163.10.3.

162A.7 Channel Operating Margin (COM)

The Channel Operating Margin (COM) for the channel between TP0 and TP5, computed using the procedure in 93A.1 and the parameters in Table 163–11, is recommended to be greater than or equal to 3 dB.